EXPERIMENT-2

Aim: To convert the ER diagram of an Indian E-commerce platform into a relational schema and implement it, using MYSQL with Primary Key, Foreign Key, NOT NULL, UNIQUE, ON DELETE CASCADE and SET NULL constraints.

Implementation:

```
Create Table Customer (
        customer_id INT PRIMARY KEY,
        name VARCHAR(50) Not NULL,
        email VARCHAR(100) UNIQUE,
        phone VARCHAR(15));
Create Table Seller (
        seller_id INT PRIMARY KEY,
        name VARCHAR(50) NOT NULL, );
Create Table Category (
        category_id INT PRIMARY KEY,
        name VARCHAR(50) UNIQUE NOT NULL);
Create Table product (
        product_id INT PRIMARY KEY,
        name VARCHAR NOT NULL,
        price DECIMAL(10,2),
        category_id INT,
        seller_id INT,
        FOREIGN KEY (category_id) REFERENCES category(category_id)
                        ON DELETE SET NULL,
        FOREIGN KEY (seller_id) REFERENCES Seller (seller_id)
                        ON DELETE SET NULL, );

Create Table Orders (
        order_id INT PRIMARY KEY,
        customer_id INT NOT NULL,
        order_date DATE,
        total_amount DECIMAL(10,2),
        FOREIGN KEY (customer_id) REFERENCES Customer (customer_id)
                        ON DELETE CASCADE );

Create Table OrderItems (
        order_id INT,
        product_id INT,
        Quantity INT NOT NULL,
        PRIMARY KEY (order_id, product_id),
        FOREIGN KEY (order_id) REFERENCES Order (order_id)
                ON DELETE CASCADE,
        FOREIGN KEY (product_id) REFERENCES product (product_id)
                ON DELETE CASCADE);

create Table Payment (
        payment_id INT PRIMARY KEY,
        order_id INT UNIQUE,
        amount DECIMAL(10,2),
        FOREIGN KEY order_id REFERENCES Orders (order_id)
                        ON DELETE CASCADE );
```

```
create Table Delivery (
        delivery_id INT PRIMARY KEY,
        status varCHAR(30),
        FOREIGN KEY (order_id) REFERENCES Order(order_id)
                ON DELETE CASCADE );
```

Sample data:

```
INSERT INTO Customer values
(1,'Rahul','rahul@gmail.com','9876543210');
Insert into Seller Values
(101,'Techstore');
Insert into category values
(201,'Electronics');
Insert into Product Values
(301,'Laptop', 65000,201,101);
Insert into Order values
(401,1,'2026-08-20', 65000);
Insert into OrderItem Values
(401,301,1);
Insert into Payment Values
(501,401,65000);
Insert into Delivery values
(601,401,'Processing');
```

Integrity Test:

```
-- INVALID CustomerID → foreign key violation
Insert into Order Values
(402,99,'2026-08-20',5000);
```

Output:

1 row successfully inserted into Customers.
1 row successfully inserted into Seller.
1 row successfully inserted into Category.
1 row successfully inserted into Product.
1 row successfully inserted into Orders.
1 row successfully inserted into OrderItems.
1 row successfully inserted into Payment.
1 row successfully inserted into Delivery.
All tables created & sample data inserted successfully.

☒ ERROR 1452 (23000):
Cannot add or update a child row:
a foreign key constraint fails.

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